GLOBAL COMPARATIVE ANALYSIS

Global Life Expectancy Paradox

A Comparative Analysis of Country-Level Data: Initial Findings and Insights from the Cleaned Dataset



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PURPOSE

Aim of The Analysis

This study aims to examine the differences in life expectancy across countries by analysing a range of economic, healthcare, educational, social, demographic, technological, and environmental indicators. The analysis begins with an assessment of the original dataset, followed by an investigation of the cleaned dataset to identify meaningful patterns and relationships. The primary objective is to determine the extent to which a country's economic wealth influences life expectancy and to explore the role of other factors that may contribute to differences in longevity.

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Initial Review of the Original Dataset

My initial step involved assessing the size, coverage, and overall quality of the original dataset. The preliminary data quality checks revealed several inconsistencies and missing values, highlighting the need for data cleaning before conducting reliable comparisons of life expectancy across countries.

What I checked	Finding
Rows and columns	204 rows × 38 columns
Distinct country names	192
Exact duplicate rows	12
Blank cells	305
Regions represented	22

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Missing Data in the Original Dataset

The missing values are unevenly distributed across the indicators, with certain variables containing more gaps than others. This is important because indicators with a higher proportion of missing data may have fewer countries available for comparison, potentially affecting the consistency and reliability of the analysis.

Indicator	Missing values
co2	59
Post-secondary enrollment (female)	33
Post-secondary enrollment (male)	33
Homicide rate	23
Refugees	14

HIGHEST OBSERVED GAP

59

co2 missing values

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From the Original Dataset to the Cleaned Dataset

After removing 12 exact duplicate rows, the dataset was reduced from 204 to 192 country records. The cleaned dataset used in my analysis contains 20 selected variables, including male and female life expectancy. It also incorporates key indicators such as GDP per capita, infant mortality, fertility rate, internet usage, school enrollment, unemployment, urban population, forested area, and co2 emissions. This refined dataset provides a more consistent basis for comparing countries and examining factors associated with life expectancy. However, careful consideration of missing values and the accuracy of country identifiers remains essential to ensure reliable results.



204 rows × 38 columns



192 country records



20 selected variables

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What Life Expectancy Looks Like

The analysis of the cleaned dataset indicates that female life expectancy is higher than male life expectancy in terms of both the mean and median values. However, considerable variation exists across countries, highlighting the differences in life expectancy between populations and the need to examine the factors that may contribute to these disparities.

Measure	Male life expectancy	Female life expectancy
Mean	69.64 years	74.51 years
Median	70.55 years	76.55 years
Minimum	50.4 years	54.8 years
Maximum	81.7 years	87.5 years

MEAN GAP

4.87

years higher for female life expectancy

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The Paradox: Wealth Is Not the Whole Story

GDP per capita shows considerable variation across countries, with a mean of approximately US$15,500 and a median of around US$5,900. This difference suggests that a relatively small number of high-income countries raise the average, making the median a more representative measure of the typical country's economic position. Although economic resources may influence life expectancy, they do not provide a complete explanation for differences in longevity. Factors such as healthcare conditions, infant mortality, education, fertility, technological access, and overall living standards must also be considered. Therefore, my analysis examines these indicators collectively to develop a broader understanding of the global life expectancy paradox, rather than attributing differences in life expectancy to economic wealth alone.

US$15,500

mean

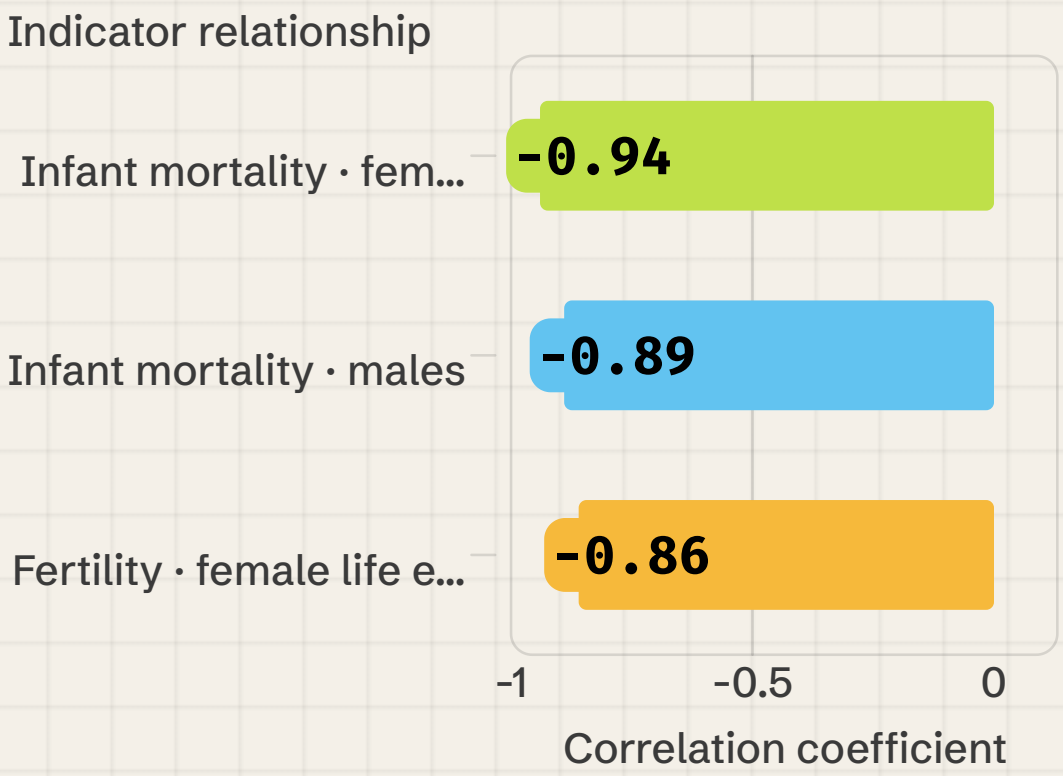
US$5,900

median

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Health and Social Indicators Show Strong Links

The correlation analysis reveals a strong negative relationship between infant mortality and life expectancy, with correlation coefficients of −0.94 for females and −0.89 for males. Female life expectancy also exhibits a strong negative correlation with fertility, with a coefficient of −0.86. These findings indicate that countries with higher infant mortality and fertility rates tend to have lower life expectancy. The observed relationships highlight the importance of considering broader health and demographic indicators when investigating differences in life expectancy across countries. However, these correlations indicate associations rather than direct causal relationships.



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Technology, Education and Living Conditions

Internet usage and school enrollment offer additional perspectives for examining differences in development across countries. The correlation analysis shows a strong positive relationship between male and female school enrollment (0.97), as well as between male and female life expectancy (0.96). These strong associations suggest that related indicators may reflect overlapping underlying patterns rather than entirely independent influences. Therefore, my analysis considers these variables alongside healthcare indicators and GDP per capita to gain a more comprehensive understanding of differences in life expectancy across countries.

SCHOOL ENROLLMENT

0.97

male and female school enrollment

LIFE EXPECTANCY

0.96

male and female life expectancy

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Next steps

1. My next step is to examine the relationship between life expectancy and key indicators, including GDP per capita, infant mortality, fertility, internet usage, and school enrollment.

2. I will investigate whether countries with comparable GDP per capita levels exhibit substantial differences in life expectancy, while also examining regional patterns and the number of countries represented in each region.

3. Where possible, I will repeat the key comparisons using both imputed and non-imputed values to assess the impact of missing-data treatment on the findings.

4. Throughout the analysis, I will acknowledge limitations related to data availability, regional representation, and the interpretation of correlations, ensuring that the conclusions remain balanced and evidence-based.

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